

Formale Methoden der Informatik

Tutorial: Induction Proofs

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Informatics

Outline

Induction Principles

Mathematical Induction

Principle of Strong (Mathematical) Induction

Principle of Structural Induction

Noetherian (or Well-founded) Induction

Hints to Write Down Induction Proofs

Learning Objectives

How can we prove the following statements?

- Summation formulas like $\sum_{i=0}^n 2^i = 2^{n+1} - 1$ for all $n \in \mathbb{N}_0$
- Inequalities like $2^n < n!$ for each integer $n \geq 4$
- Divisibility results like $n^3 - n$ is divisible by 3 for each $n \in \mathbb{N}$
- Results about sets like any set of $n \in \mathbb{N}$ elements has 2^n subsets
- Results about algorithms like correctness or termination
The function `fac(n)` returns $n!$ for all $n \in \mathbb{N}_0$

Induction Principles

- A central proof techniques in math and computer science:

Induction principles

- Formalized as an axiom schema like

$$[\mathcal{P}(1) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))] \rightarrow \forall n \in \mathbb{N} \mathcal{P}(n)$$

- Different kinds of induction
 - Mathematical induction (and its variants)
 - Strong mathematical induction
 - Structural induction (e.g., for inductively defined data types)
 - Noetherian (or well-founded) induction
- Details and examples are presented later.

How does it work behind the scene?

$$[\mathcal{P}(1) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))] \rightarrow \forall n \in \mathbb{N} \mathcal{P}(n) \quad (\text{MI})$$

We have to show $\mathcal{P}(1) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))$ because then $\forall n \in \mathbb{N} \mathcal{P}(n)$ can be derived with (MI) by modus ponens (MP).

β , the proof of $\forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))$, is obtained as follows:

1. Take an arbitrary $\ell \in \mathbb{N}$ (an “eigenvariable”) and
2. assume $\mathcal{P}(\ell)$ and derive $\mathcal{P}(\ell+1)$.

Then we have

$$\frac{\frac{\mathcal{P}(1)}{\alpha} \quad \frac{\forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))}{\beta}}{\mathcal{P}(1) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k) \rightarrow \mathcal{P}(k+1))} \wedge I \quad \frac{}{\forall n \in \mathbb{N} \mathcal{P}(n)} (\text{MI}) \quad \text{MP}$$

Principle of mathematical induction

Let $\mathcal{P}(n)$ be a statement involving a variable n . Suppose

1. $\mathcal{P}(1)$ is true;
2. if $\mathcal{P}(k)$ is true for some natural numbers $k \geq 1$, then $\mathcal{P}(k+1)$ is also true.

Then $\mathcal{P}(n)$ is true for all natural numbers $n = 1, 2, \dots$

- In the **base case**, we prove that $\mathcal{P}(1)$ is true.
- We are allowed to assume that $\mathcal{P}(k)$ is true for some $k \geq 1$. This is the **induction hypothesis**.
- We prove $\mathcal{P}(k+1)$ true in the **induction step**. In the course of this proof, we usually apply the induction hypothesis.

Principle of mathematical induction

An example proof using mathematical induction

Theorem: The sum of the first n positive odd integers is n^2 .

Proof. We want to prove that, for all natural numbers n , $\mathcal{P}(n)$ holds, where $\mathcal{P}(n)$ denotes $\sum_{i=0}^{n-1} (2i + 1) = n^2$. The proof is by mathematical induction on n .

Base case. We show that $\mathcal{P}(1)$ is true. This is the case since

$$\sum_{i=0}^{n-1} (2i + 1) = 1 = n^2.$$

Induction hypothesis. Assume $\mathcal{P}(k)$ is true for some $k \geq 1$.

Induction step. We want to show that $\mathcal{P}(k + 1)$ is true. Therefore, we need to show that $\sum_{i=0}^k (2i + 1) = (k + 1)^2$ holds.

Principle of mathematical induction

An example proof using mathematical induction (cont'd)

By the induction hypothesis, $\mathcal{P}(k)$ is true. Then we derive

$\mathcal{P}(k)$ is true iff

$$\sum_{i=0}^{k-1} (2i+1) = k^2 \quad \text{iff}$$

$$2k+1 + \sum_{i=0}^{k-1} (2i+1) = k^2 + 2k + 1 \quad \text{iff}$$

$$\sum_{i=0}^k (2i+1) = (k+1)^2.$$

Hence, $\mathcal{P}(k+1)$ is true.



What if more than one variable is involved?

Sometimes additional (free) variables are required (examples below)

How can we handle them?

We consider 3 approaches (for different circumstances)

1. Universal variables as additional free variables of $\mathcal{P}$
2. Universal quantifiers inside $\mathcal{P}$
3. Induction over tuples (➡ Noetherian induction discussed later)

Universal variables as additional free variables of $\mathcal{P}$

Correctness of a more general induction principle (MI $\forall$)

$$\begin{aligned} \forall x \left[(\mathcal{P}(1, x) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k, x) \rightarrow \mathcal{P}(k+1, x))) \right] \\ \rightarrow \forall x \forall n \in \mathbb{N} \mathcal{P}(n, x) \end{aligned} \quad (\text{MI}\forall)$$

Theorem: (MI $\forall$) holds.

Proof (sketch) by contradiction. Assume that (MI $\forall$) is false. Then the left hand side (LHS) of the implication is true and the right HS (RHS) $\forall x \forall n \in \mathbb{N} \mathcal{P}(n, x)$ is false. This implies that we have a smallest n_0 such that $\mathcal{P}(n_0, x_0)$ is false, $n_0 \geq 2$ because $\mathcal{P}(1, x_0)$ is true, and $\mathcal{P}(n_0 - 1, x_0)$ is true. Since $\mathcal{P}(n_0 - 1, x_0) \rightarrow \mathcal{P}(n_0, x_0)$ is true by assumption, the truth of $\mathcal{P}(n_0, x_0)$ follows. Contradiction!

Universal variables as additional free variables of $\mathcal{P}$

The more general induction principle (MI $\forall$) in practice

$$\begin{aligned} \forall x \left[(\mathcal{P}(1, x) \wedge \forall k \in \mathbb{N} (\mathcal{P}(k, x) \rightarrow \mathcal{P}(k+1, x))) \right] \\ \rightarrow \forall x \forall n \in \mathbb{N} \mathcal{P}(n, x) \end{aligned} \quad (\text{MI}\forall)$$

In practice:

- Fix x to an arbitrary allowed value a (a constant!) before the induction proof
- Then the LHS of (MI) is obtained from the LHS of (MI $\forall$) and
- we can proceed by (MI) as before
- Attention: Sometimes it is necessary to have quantifiers in $\mathcal{P}$!

Principle of mathematical induction

A correctness and termination proof by mathematical induction

Algorithm 1: The power function $pow(b, n): \mathbb{N} \times \mathbb{N}_0 \mapsto \mathbb{N}$

Input: b, n : a positive integer and a non-negative integer

Output: The computed positive integer value for b, n

```
1 if  $n == 0$  then return 1 ;  
2 else return  $b \times pow(b, n - 1)$  ;
```

Prove correctness of $pow(\cdot, \cdot)$ (i.e., $pow(b, n) = b^n$) using (MI $\forall$).

Proof. We want to prove that, for all positive integers b and all natural numbers n (including 0), $\mathcal{P}(n, b)$ holds, where $\mathcal{P}(n, b)$ denotes the statement that **for the positive integer b and the non-negative integer n , $pow(b, n)$ computes b^n** . The proof is by **mathematical induction on n using (MI $\forall$)**.

Principle of mathematical induction

A correctness and termination proof by mathematical induction (cont'd)

Base case. We show that $\mathcal{P}(0, b)$ is true. This is the case since $\text{pow}(b, 0)$ terminates in line 1 and returns the correct value $b^0 = 1$ for the positive integer b .

Induction hypothesis. Assume $\mathcal{P}(k, b)$ is true for some integer $k \geq 0$.

Induction step. We want to show that $\mathcal{P}(k + 1, b)$ is true. Consider $\text{pow}(b, k + 1)$. Since $k + 1 > 0$, line 2 is reached and $b \times \text{pow}(b, k)$ is computed. From the induction hypothesis, we know that $\text{pow}(b, k)$ computes b^k for the positive integer b . Thus $\text{pow}(b, k + 1)$ computes the correct value $b \times b^k = b^{k+1}$. Hence, $\mathcal{P}(k + 1, b)$ is true. $\square$

Remark: We started the induction from 0. What do we have to change in order to use exactly the mathematical induction schema from above?

Principle of mathematical induction

An example with universal quantifiers inside $\mathcal{P}$

Show for all non-negative integers m , k , and n :

$$\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} \quad (\text{B})$$

Let $\mathcal{P}(m)$ denote the statement that (B) is true for all non-negative integers k and n . The proof is by mathematical induction on m .

Base case: $m = 0$. We show that $\mathcal{P}(0)$ is true. Choose two arbitrary non-negative integers k and n . Then, with $\binom{m}{i} = 0$ if $i > m$,

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \sum_{i=0}^k \binom{0}{i} \binom{n}{k-i} = \binom{0}{0} \binom{n}{k} = \binom{n}{k} = \binom{m+n}{k}$$

Principle of mathematical induction

An example with universal quantifiers inside $\mathcal{P}$

Induction hypothesis. Assume $\mathcal{P}(m)$ is true for some integer $m \geq 0$.

Induction step. We want to show that $\mathcal{P}(m+1)$ is true. Choose two arbitrary non-negative integers k and n .

$$\begin{aligned}\sum_{i=0}^k \binom{m+1}{i} \binom{n}{k-i} &= \sum_{i=1}^k \binom{m+1}{i} \binom{n}{k-i} + \binom{n}{k} \\&= \sum_{i=1}^k \left[\binom{m}{i} + \binom{m}{i-1} \right] \binom{n}{k-i} + \binom{n}{k} \\&= \sum_{i=1}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i} + \binom{n}{k} \\&= \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i}\end{aligned}$$

Principle of mathematical induction

An example with universal quantifiers inside $\mathcal{P}$

$$\begin{aligned}\sum_{i=0}^k \binom{m+1}{i} \binom{n}{k-i} &= \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i} \\ &= \sum_{i=0}^{\textcolor{red}{k}} \binom{m}{i} \binom{n}{\textcolor{red}{k}-i} + \underbrace{\sum_{i=0}^{\textcolor{red}{k}-1} \binom{m}{i} \binom{n}{\textcolor{red}{k}-1-i}}_{=0 \text{ by definition}}\end{aligned}$$

Case 1: $k = 0$: Then the second sum is zero. We get

$$\begin{aligned}\sum_{i=0}^k \binom{m+1}{i} \binom{n}{k-i} &= \sum_{i=0}^{\textcolor{red}{k}} \binom{m}{i} \binom{n}{\textcolor{red}{k}-i} + \underbrace{\sum_{i=0}^{\textcolor{red}{k}-1} \binom{m}{i} \binom{n}{\textcolor{red}{k}-1-i}}_{=0 \text{ by definition}} \\ &= \sum_{i=0}^{\textcolor{red}{k}} \binom{m}{i} \binom{n}{\textcolor{red}{k}-i} = \binom{m}{0} \binom{n}{0} = 1 = \binom{m+1+n}{k}\end{aligned}$$

Principle of mathematical induction

An example with universal quantifiers inside $\mathcal{P}$

Case 2: $k > 0$: We get

$$\begin{aligned}\sum_{i=0}^k \binom{m+1}{i} \binom{n}{k-i} &= \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i} \\&= \sum_{i=0}^{\textcolor{red}{k}} \binom{m}{i} \binom{n}{\textcolor{red}{k}-i} + \sum_{i=0}^{\textcolor{red}{k}-1} \binom{m}{i} \binom{n}{\textcolor{red}{k}-1-i} \\&=_{\text{IH}} \binom{m+n}{k} + \binom{m+n}{k-1} = \binom{m+1+n}{k}\end{aligned}$$

Hence $\mathcal{P}(m+1)$ is true. □

Remark We apply the induction hypothesis twice, once to a binomial coefficient where $\textcolor{red}{k}$ is involved and once to one with $\textcolor{red}{k}-1$. This is the reason why we need the (strong) induction hypothesis with the universal quantifiers inside $\mathcal{P}$.

The handling of the universal quantifiers inside $\mathcal{P}$

In the base case: Show the universal statement $\mathcal{P}(0)$, i.e., show it for arbitrary non-negative integers k and n .

When the induction hypothesis is used: You are allowed to use any legal instantiation of the universal quantifiers inside $\mathcal{P}(m)$.

In the induction step: Show the universal statement $\mathcal{P}(m+1)$, i.e., show it for arbitrary non-negative integers k and n .

Principle of mathematical induction: Variation 1

Let $\mathcal{P}(n)$ be a statement involving a variable n . Suppose

1. $\mathcal{P}(k_0)$ is true for some natural number k_0 ;
2. if $\mathcal{P}(k)$ is true for some natural numbers $k \geq k_0$, then $\mathcal{P}(k+1)$ is also true.

Then $\mathcal{P}(n)$ is true for all natural numbers $n = k_0, k_0 + 1, \dots$

Useful, e.g., to prove that $2^n > n^2$ for all natural numbers $n \geq 5$.

Remark: Variation 1 can be translated to standard mathematical induction by $\mathcal{Q}(r - k_0 + 1) = \mathcal{P}(r)$, where $k_0 \leq r$.

Principle of mathematical induction: Variation 1

Prove: $2^n > n^2$ for all natural numbers $n \geq 5$.

Proof. Let $\mathcal{P}(n)$ denote the statement $2^n > n^2$. The proof is by mathematical induction on n starting with $n_0 = 5$.

Base case: $n_0 = 5$. $\mathcal{P}(n_0)$ is true because $2^5 = 32 > 25 = 5^2$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq n_0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. By the induction hypothesis, $\mathcal{P}(n)$ is true. Then we derive

$$\begin{aligned} 2^n &> n^2 \quad \text{iff} \\ 2 \cdot 2^n &> 2 \cdot n^2 \quad \text{iff} \\ 2^{n+1} &> 2 \cdot n^2. \end{aligned}$$

Since $n \geq 5$, $2 \cdot n^2 > (n+1)^2$ holds. Then $2^{n+1} > (n+1)^2$ follows by the transitivity of $>$. Hence, $\mathcal{P}(n+1)$ is true. $\square$

Principle of mathematical induction: Variation 2

Its principle and a short example

Let $\mathcal{P}(n)$ be a statement involving a variable n . Suppose

1. $\mathcal{P}(1)$ and $\mathcal{P}(2)$ are true;
2. if $\mathcal{P}(k)$ and $\mathcal{P}(k+1)$ are true for some natural number $k \geq 1$, then $\mathcal{P}(k+2)$ is also true.

Then $\mathcal{P}(n)$ is true for all natural numbers $n = 1, 2, \dots$

Further generalizations are possible: Use more than two levels, start with some $n > 1$, etc.

Useful, e.g., to prove the following:

Let $\{a_n\}$ be a sequence of natural numbers such that $a_1 = 5$, $a_2 = 13$ and $a_{n+2} = 5a_{n+1} - 6a_n$ for all natural numbers n . Show that $a_n = 2^n + 3^n$ holds for all natural numbers n .

Principle of mathematical induction: Variation 2

A proof for the example

Proof. We want to prove that, for all natural numbers n , $\mathcal{P}(n)$ holds, where $\mathcal{P}(n)$ denotes $a_n = 2^n + 3^n$. The proof is by variant 2 of mathematical induction on n .

Base cases. $\mathcal{P}(1)$ is true, because $a_1 = 5 = 2^1 + 3^1$. Moreover, $\mathcal{P}(2)$ is true, because $a_2 = 13 = 2^2 + 3^2$.

Induction hypothesis. Assume $\mathcal{P}(k)$ and $\mathcal{P}(k+1)$ are true for some $k \geq 1$.

Induction step. We want to show that $\mathcal{P}(k+2)$ is true.

Principle of mathematical induction: Variation 2

A proof for the example (cont'd)

From the induction hypothesis, we have $a_k = 2^k + 3^k$ and $a_{k+1} = 2^{k+1} + 3^{k+1}$. We compute

$$\begin{aligned} a_{k+2} &= 5 \cdot a_{k+1} - 6 \cdot a_k \\ &= 5 \cdot 2^{k+1} + 5 \cdot 3^{k+1} - 6 \cdot 2^k - 6 \cdot 3^k \\ &= 5 \cdot 2^{k+1} + 5 \cdot 3^{k+1} - 3 \cdot 2^{k+1} - 2 \cdot 3^{k+1} \\ &= 2 \cdot 2^{k+1} + 3 \cdot 3^{k+1} \\ &= 2^{k+2} + 3^{k+2} \end{aligned}$$

Hence, $\mathcal{P}(k+2)$ is true. □

We obtained a **closed form** for an inductive/recursive definition.

Where is this applied in computer science?

Principle of strong (mathematical) induction

Let $\mathcal{P}(n)$ be a statement involving a variable n . Suppose

1. $\mathcal{P}(1)$ is true;
2. if, for some natural number $k \geq 1$, $\mathcal{P}(1), \mathcal{P}(2), \dots, \mathcal{P}(k)$ are all true, then $\mathcal{P}(k+1)$ is also true.

Then $\mathcal{P}(n)$ is true for all natural numbers $n = 1, 2, \dots$

The induction hypothesis is **stronger** (compared to above).

NB: Stronger means potentially less models

Find examples where strong induction can be beneficially applied!

Principle of structural induction

In order to show that a statement holds **for all elements of a inductively/recursively defined set**, use the following:

Base case(s). Prove that the statement holds for all elements specified in the base case(s) of the set definition.

Induction step. Prove that if the statement is true for each of the elements used to construct elements in the inductive step of the set definition, then the result holds for these new elements.

- ➡ The structure of a structural induction proof “follows” the structure of the underlying definition.

Principle of structural induction

The basic definitions for the list example

Consider the following definition of lists

$$lst ::= nil \mid (c : lst)$$

where $:$ means LISP cons and nil denotes the empty list.

Example

A list with three elements 1, 2, 3 looks as follows:

$$(1 : (2 : (3 : nil)))$$

Principle of structural induction

Appending two lists

We want to implement an **append** (*app*) function recursively:

$$app(nil, y) = y \quad (1)$$

$$app((c : x), y) = (c : app(x, y)) \quad (2)$$

Example

We want to append $(a : (b : nil))$ and $(c : nil)$

$$\begin{aligned} app((a : (b : nil)), (c : nil)) &=_{(2)} (a : app((b : nil), (c : nil))) \\ &=_{(2)} (a : (b : app(nil, (c : nil)))) \\ &=_{(1)} (a : (b : (c : nil))) \end{aligned}$$

Principle of structural induction

Proving properties about *app*

$$lst ::= nil \mid (c : lst)$$

Let $\mathcal{P}(x)$ denote $app(x, nil) = x$.

Show that, for all lists ℓ , $\mathcal{P}(\ell)$ holds.

Proof: We proceed by structural induction on the definition of lists.

Basis case: $\ell = nil$. Then $\mathcal{P}(\ell)$ is $app(nil, nil) = nil$ which follows immediately from the first defining equality (1).

Induction hypothesis. Assume that $\mathcal{P}(\ell)$ holds for some list ℓ .

Principle of structural induction

Proving properties about *app* (cont'd)

$$lst ::= nil \mid (c : lst)$$

Induction step. We have to establish $\mathcal{P}((c : \ell))$. The induction hypothesis gives us

$$app(\ell, nil) = \ell.$$

We concatenate c to the left on both sides resulting in

$$(c : app(\ell, nil)) = (c : \ell).$$

By the second defining equality (2), we get

$$app((c : \ell), nil) = (c : \ell).$$

Hence, $\mathcal{P}((c : \ell))$ holds.

Consequently, $\mathcal{P}(\ell)$ is true for all lists. □

How can we use mathematical induction to prove the above result?

Noetherian¹ (or well-founded) induction

Partially ordered sets



Definition (Partially ordered sets (posets))

A **partial order** is a binary relation $\leq$ over a set S which is

1. reflexive: $a \leq a$ holds $\forall a \in S$;
2. antisymmetric: $a \leq b \wedge b \leq a \rightarrow a = b$ holds $\forall a, b \in S$; and
3. transitive: $a \leq b \wedge b \leq c \rightarrow a \leq c$ holds $\forall a, b, c \in S$.

$(S, \leq)$ (or often simply S) is called a **partially ordered set**.

Examples: $(\mathbb{X}, \leq)$ for $\mathbb{X} \in \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}\}$

Fact: If $(S, \leq_S)$ is a poset, $T \subseteq S$, and $\leq_T$ is the restriction of $\leq_S$ to $T \times T$, then $(T, \leq_T)$ is also a poset.

A poset is **totally ordered** if every pair of elements is comparable.

¹**Emmy Noether**  (1882–1935): influential female German mathematician

Noetherian induction

Well-founded sets

Given a poset $(S, \leq)$, the **strict order** $<$ associated with $\leq$ is defined as follows:

$$x < y \quad \text{if and only if} \quad x \leq y \wedge x \neq y$$

Definition (Well-founded sets)

$(S, \leq)$ is **well-founded** iff it has no infinite descending chain, i.e., no infinite, strictly decreasing sequence of elements

$$x_1 > x_2 > \cdots > x_n > \cdots .$$

Examples: $(\mathbb{N}, \leq)$ is well-founded, but $(\mathbb{Z}, \leq)$ is not.

Noetherian induction

Well-founded sets

Given a poset $(S, \leq)$ and $X \subseteq S$.

- An element $b \in X$ is **the least element of X** iff for all $x \in X$, $b \leq x$ holds.
- An element $b \in X$ is **minimal in X** iff for all $x \in X$, $x \leq b$ implies $x = b$.
- An element $b \in X$ is **minimal in X** iff there is no $x \in X$ such that $x < b$ holds.
- Minimal elements are not necessarily unique.

Lemma

$(S, \leq)$ is **well-founded** iff every non-empty subset of S contains at least one minimal element with respect to the order relation $\leq$.

Noetherian (or well-founded) induction

Lexicographic orders

Definition (Lexicographic order)

Let $(S, \leq)$ be a poset. The **lexicographic order** $\sqsubseteq$ on $S \times S$ is given by

$$(s, t) \sqsubseteq (s', t') \quad \text{iff} \quad \begin{cases} s < s', \\ s = s' \text{ and } t \leq t' \end{cases}$$

for all s, s', t, t' in S .

Fact: If $(S, \leq)$ is a poset, then $(S \times S, \sqsubseteq)$ is a poset.

Fact: Let $(S, \leq)$ be a poset. If $(S, \leq)$ is well-founded, then the lexicographically ordered set $(S \times S, \sqsubseteq)$ is also well-founded.

Noetherian (or well-founded) induction

The principle

Let $(S, \leq)$ be a well-founded set and let $\mathcal{P}(x)$ be a statement involving a variable x . Suppose

1. $\mathcal{P}(m)$ is true for each minimal element m of S ;
2. for each non-minimal element x , if $\mathcal{P}(y)$ is true for all $y < x$, then $\mathcal{P}(x)$ is also true.

Then $\mathcal{P}(x)$ is true for all $x \in S$.

The schema (NI) for Noetherian induction is as follows:

$$\forall x \in S [(\forall y \in S (y < x \rightarrow \mathcal{P}(y))) \rightarrow \mathcal{P}(x)] \rightarrow \forall z \in S \mathcal{P}(z) \quad (\text{NI})$$

Noetherian (or well-founded) induction

The principle

$$\forall x \in S \left[\left(\forall y \in S (y < x \rightarrow \mathcal{P}(y)) \right) \rightarrow \mathcal{P}(x) \right] \rightarrow \forall z \in S \mathcal{P}(z) \quad (\text{NI})$$

Correctness of the Noetherian induction principle

Theorem: (NI) holds for any well-founded set.

Proof by contradiction. Assume that (NI) is false. Then

$$\forall x \in S [(\forall y \in S (y < x \rightarrow \mathcal{P}(y))) \rightarrow \mathcal{P}(x)] \quad (3)$$

is true and

$$\forall z \in S \mathcal{P}(z)$$

is false, i.e., $\exists z \in S \neg \mathcal{P}(z)$ holds. Let $X = \{x \in S \mid \neg \mathcal{P}(x)\}$. X is non-empty by construction. Since S is well-founded, X has some minimal element m (by the lemma above). Instantiating x in (3) by m , we get that

$$(\forall y \in S (y < m \rightarrow \mathcal{P}(y))) \rightarrow \mathcal{P}(m) \quad (4)$$

is true. We consider the following two cases.

Correctness of the Noetherian induction principle (cont'd)

Case 1: m is minimal in S . Then there is no $y \in S$ s.t. $y < m$ (why?) and

$$\forall y \in S (y < m \rightarrow \mathcal{P}(y)) \quad (5)$$

holds trivially. But then the truth of (4) implies that $\mathcal{P}(m)$ is true which contradicts $m \in X$.

Case 2: m is not minimal in S . Then, for all $y \in S$ with $y < m$, $\mathcal{P}(y)$ has to be true, since otherwise $y \in X$ and m cannot be minimal in X . But then (5) holds. The rest is as in Case 1.

Hence, (NI) is valid for well-founded sets. □

The answer to the why question on the last slide

If m is minimal in S then $\neg \exists y \in S (y < m)$

m minimal in S	iff	$\forall y \in S (y \leq m \rightarrow y = m)$	Def. minimal ele
	iff	$\neg \neg \forall y \in S (y \leq m \rightarrow y = m)$	$\neg \neg F \equiv F$
	iff	$\neg \exists y \in S (y \leq m \wedge y \neq m)$	De Morgan
	iff	$\neg \exists y \in S (y < m)$	Def. $<$

The Ackermann-Péter² function $A(x, y): \mathbb{N}_0 \times \mathbb{N}_0 \mapsto \mathbb{N}_0$

Definition of the function

Algorithm 2: Ackermann-Péter function

Input: x, y , two non-negative integers

Output: The computed non-negative integer value for x, y

```
1 if  $x == 0$  then  
2   return  $y + 1$ ;  
3 else if  $y == 0$  then  
4   return  $A(x - 1, 1)$ ;  
5 else return  $A(x - 1, A(x, y - 1))$ ;
```

- This function is well known for its extraordinary growth.
- Try it out by hand or implement it.
- Does the function terminate for all admissible inputs x and y and return a non-negative integer?

²Wilhelm A.  (1896–1962) German mathematician, student of Hilbert 
Rózsa P.  (1905–1977) Hungarian mathematician, mother of recursion theory

The Ackermann-Péter function

The termination proof

We know: $(\mathbb{N}_0 \times \mathbb{N}_0, \sqsubseteq)$ is well-founded with least element $(0, 0)$.

Let $\mathcal{P}(x, y)$ denote the statement

“ $A(x, y)$ terminates on inputs $(x, y) \in \mathbb{N}_0 \times \mathbb{N}_0$ and returns a non-negative integer value”.

The proof is by Noetherian induction on $(\mathbb{N}_0 \times \mathbb{N}_0, \sqsubseteq)$.

Base case: The least element is $(0, 0)$. Then $\mathcal{P}(0, 0)$ is true because $A(0, 0) = 1$ (line 2).

Induction hypothesis. Pick arbitrarily a non-minimal element (x, y) and assume that $\mathcal{P}(x', y')$ is true for all $(x', y') \sqsubset (x, y)$.

Induction step. We want to show that $\mathcal{P}(x, y)$ is true. Recall that $x \geq 0$ and $y \geq 0$. We distinguish the following three cases.

The Ackermann-Péter function

The termination proof (cont'd)

Case 1: $x = 0$. Then $A(0, y) = y + 1$ (line 2) and $\mathcal{P}(0, y)$ is true.

Case 2: $x \neq 0 \wedge y = 0$. Then $(x - 1, 1) \sqsubset (x, 0)$. By the induction hypothesis, $\mathcal{P}(x - 1, 1)$ is true and $A(x, 0) = A(x - 1, 1)$ (line 4). Therefore, $\mathcal{P}(x, 0)$ is true.

Case 3: $x \neq 0 \wedge y \neq 0$. Then $(x, y - 1) \sqsubset (x, y)$ and, for all $z \in \mathbb{N}_0$, $(x - 1, z) \sqsubset (x, y)$. Then, $\mathcal{P}(x, y - 1)$ is true by the induction hypothesis. Moreover, for all $z \in \mathbb{N}_0$, $\mathcal{P}(x - 1, z)$ is also true by the induction hypothesis. $A(x, y) = A(x - 1, A(x, y - 1))$ (line 5) therefore terminates and computes a non-negative integer value. Hence, $\mathcal{P}(x, y)$ is true.

We conclude that $\mathcal{P}(x, y)$ is true for all $(x, y) \in \mathbb{N}_0 \times \mathbb{N}_0$. □

Induction principles not covered in this lecture

Reverse induction 

Backward induction 

Cauchy induction 

Mutual induction

Hopcroft, J. E., Motwani, R. and Ullman, J. D.: *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Pearson, 2007

How to write down an induction proof?

- Identify the claim you are going to prove.
- Let the reader know which kind of induction you will use.
- For structural induction, a claim is about all elements of some inductively defined set. It is a good idea to indicate the inductive definition on which the proof is based.
- Make clear (e.g., by labels) what the basis, the induction hypothesis and the induction step are.
- State precisely the induction hypothesis you are going to use and indicate the claim you prove in the induction step.
- Make clear where you apply the induction hypothesis.
- Check whether I followed the advise! Report violations!

Learning objectives

- Ability to discuss different induction principles in detail and to distinguish between them.
- Ability to apply these principles and to construct proofs by induction.